

UTM CSC343: Introduction to Databases
Midterm Examination October 25, 2002 12:10-1:00

Student Number: _____

Last Name: _____ First Name: _____
This exam consists of 3 questions on 4 double sided pages. Unless otherwise specified, assume that you are using a SQL standard system (ie that has assertions). You are allowed one $8\frac{1}{2} \times 11$ single sided aid sheet. **You can find the MFSchema from Assignment 1 and the GraphSchema at the end of this exam.**

1. (12 Marks: Mixed)

This question concerns the Graph Schema **at the end of the exam** which can be used to store information about a collection of graphs. Each graph is represented by a single entry in the graph relation. The directed edges for a particular graphId are captured by a collection of $(eFrom, eTo)$ pairs for the graphId in the edge relation.

Rewrite the Graph Schema (in SQL) so that it captures the following constraints:

- (a) graphId is the unique identifier for each graph
- (b) The graphId's in the edge relation must be associated with a graphId in the graph relation
- (c) numVertices is at least 0.
- (d) If a graph is deleted, then so are all its edges.
- (e) An edge can appear at most one time in a given graph. That is, (5,2,3) can not appear twice in the edge relation.
- (f) No loops are allowed. That is, no tuples of the form (5,19,19) are allowed in the edge relation. This tuple indicates **in graph 5, there is an edge from vertex 19 to vertex 19.**

2. (20 Marks: Mixed) Using MFSchema or Graph Schema write RA or SQL as required:

- (a) **Using SQL** Find the sids of customers that have Zubair as their rep or have some transaction involving the **Precious Metals** fund.

- (b) **Using SQL** A graph is undirected if, whenever (i, j) is an edge in the graph, so is (j, i) . For example, graph 9 is undirected, graph 21 is not undirected. Write a single insert statement that undirects the graph with graphId=72. **Note:** Do not attempt to insert an edge that already exists.

(c) **Using RA** List all edges which are part of graphs containing exactly 15 vertices. The list should have schema (*graphId,eFrom,eTo*).

(d) **Using RA** Express the constraint that Atif is the only Rep that sells the **Energy** fund.

3. (10 Marks: Mixed)

(a) Draw an ER diagram capturing the situation below. **Please create entities for the InsuranceCompany as well as InsurancePolicy.**

The Ministry of Transportation Ontario (MTO  Ontario) maintains information about drivers and the accidents they are involved in. Each driver is identified by their drivers license number (dln) and has a name and address and expiry date. Each driver has an insurance policy which is identified by the insurance company and policy number and has an expiry date. Insurance companies are identified by their name, each has a head office phone number and address. Drivers can own one or more cars, but each car is owned by exactly one driver. Each car has a make, model and year and is identified by its vehicle identification number (vin). MTO always knows the **current** owner (exactly one driver) of each car. For our purposes, MTO will not track previous owners of a car. For each accident, MTO maintains a single report (no matter the number of cars/drivers involved) with a unique report id. The report captures the location, date, description for the accident. MTO also knows which cars and drivers participated in the accident as well as the damage amount (in dollars) done to each of the cars involved.

(b) Write a relational schema (table names, attribute names, keys etc.) corresponding to your ER diagram. Best marks for the fewest relations that most accurately captures the situation on the previous page. Do not write SQL CREATE TABLE statements. Make sure that foreign keys are clearly identified.

I will get you started...

InsuranceCompany(icName, phone, address)

InsurancePolicy(

(blank page for rough work)

The Graph Schema A graph consists of a collection of vertices and edges between them. The graph with Vertex set {1,2,3,4,5} and Edge set

{(1,4),(3,4),(3,2),(2,3),(1,2)}

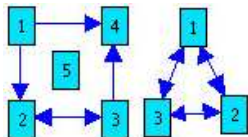
is pictured on the left below. The edge (1,4) indicates that one can get **from** vertex 1 **to** vertex 4 (by following that edge). You can think of edge (i,j) as indicating a one way street **from** i **to** j . In the above graph, you can not get from 4 to 1. You can get from 2 to 4, but not directly (as (2,4) is not an edge in the graph. To get from 2 to 4 follow edge (2,3) and then (3,4).

```
CREATE TABLE graph (
    graphId INT,
    numVertices INT
);
CREATE TABLE edge (
    graphId INT,
    eFrom INT,
    eTo INT
);
```

The following creates two graphs, a 5 vertex graph with graphId=21 and a 3 vertex undirected graph with graphId=9. The pictures of these two graphs can be found below.

```
INSERT INTO graph VALUES(21,5);
INSERT INTO edge VALUES(21,1,4);
INSERT INTO edge VALUES(21,3,4);
INSERT INTO edge VALUES(21,3,2);
INSERT INTO edge VALUES(21,2,3);
INSERT INTO edge VALUES(21,1,2);
```

```
INSERT INTO graph VALUES(9,3);
INSERT INTO edge VALUES(9,1,2);
INSERT INTO edge VALUES(9,2,1);
INSERT INTO edge VALUES(9,3,2);
INSERT INTO edge VALUES(9,2,3);
INSERT INTO edge VALUES(9,1,3);
INSERT INTO edge VALUES(9,3,1);
```



The MFSchema

```
CREATE TABLE fund(
    fundName VARCHAR(30),
    fundID INT,
    totalShares INT,
    PRIMARY KEY (fundID)
);
CREATE TABLE price(
    fundID INT,
    priceDate DATE,
    price REAL,
    PRIMARY KEY (fundID, priceDate)
);
CREATE TABLE customer (
    sin CHAR(9) PRIMARY KEY,
    name VARCHAR(20),
    address VARCHAR(20),
    phone VARCHAR(20)
);
CREATE TABLE trans(
    sin CHAR(9),
    fundID INT,
    transDate DATE,
    shares INT
);
CREATE TABLE rep(
    repid INT PRIMARY KEY,
    repName VARCHAR(20)
);
CREATE TABLE customerRep (
    sin CHAR(9),
    repId INT,
    PRIMARY KEY (sin)
);
```